# Attractive-repulsive transition of the Casimir force between anisotropic plates

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The quantized surface mode technique is used to calculate the Casimir force between two parallel anisotropic metamaterial plates with in-plan optical axes, and the main concern is focused on the possible attractive-repulsive transition of the force. The result shows that if the electric and magnetic responses of the plates are frequency dependent, the Casimir force between them is attractive when the separation is small and is able to become repulsive as the separation increases. Moreover, the polarity of the Casimir force depends on the degree of anisotropy near the zero-force position.

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The Casimir effect caused by boundary alternation of the zero-point electromagnetic energy is one of the most remarkable predictions of quantum electrodynamics [1–6]. Great effort has been put into both theoretical and experimental studies [7–18] on the Casimir effect owing to its fundamental significance in various fields of physics [3].

The Casimir force between conducting plates is always attractive, but may be repulsive when the magnetic property of plates is taken into account [19–21], therefore it is necessary to study further the properties of Casimir force between plates with frequency-dependent permittivity and permeability. Recently, the Casimir effect between anisotropic materials [14–16,18], especially metamaterial [17,22–26], has received considerable attentions. After the early study on van der Waals interaction between anisotropic bodies in 1970s [18], Capasso and collaborators calculated the Casimir torque between parallel birefringent plates [14]. In a following work, Shao *et al.* gave an analytical expression of the repulsive Casimir force between two birefringent plates with constant permittivity and permeability [15,16]. However, the attractive-repulsive transition cannot happen in this representation. In addition, the studies on the Casimir force between metamaterial plates show that the attractive-repulsive transition can occur when the optical axes are perpendicular to the surface of plates [17,22–24]. Moreover, the dissipation, metallic background and the anisotropy might affect the magnitude and the polarity of Casimir force as well. However, there is no clear and direct discussion reported so far about the influence of anisotropy on the Casimir force.

In our previous works [15,16], the quantized surface mode technique [3,27,28] was used to calculate the retarded Casimir torque between two birefringent plates with constant permittivity and permeability. In present work, we calculate the Casimir force between two metamaterial plates with frequency-dependent permittivity and permeability. The result shows that the force could change its polarity from attractive to repulsive while two plates' separation is small initially and then increasing continuously. Moreover, we also give the direct expression for the anisotropy dependence of the Casimir force which would be useful for discussing how the anisotropy exactly affects the polarity and magnitude of Casimir force.

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As shown in Fig. 1, two uniaxial plates (with the diameter $D$ and the thickness $d$) made of different materials are kept parallel at a distance $a$ in vacuum. The coordinate is chosen so that the $x$-$y$ plane is parallel to the plate's surface and the optical axes of the plates are along the $x$ direction. Assuming $D \gg a$ and $d \gg a$, the space can be approximately considered to be divided into three regions with corresponding permittivity tensors $\vec{\varepsilon}_0, \vec{\varepsilon}_1, \vec{\varepsilon}_2$ and permeability tensors $\vec{\mu}_0, \vec{\mu}_1, \vec{\mu}_2$, as shown in Fig. 2.

According to the quantized surface mode technique, we just need to consider the zero-point energy associated with the surface modes $q$ which are exponentially decaying if $z > a$ and $z < 0$ [3,15,16,27,28]. Since each region is homogeneous, the electric and magnetic fields can be expressed as

$$\vec{E}_q = iNe^{ik_x x + ik_y y}[a_q \vec{e}_q e^{ik_z z} e^{-i\omega t} - a_q^\dagger \vec{e}_q^* e^{-ik_z z} e^{i\omega t}], \quad (1)$$

$$\vec{H}_q = Ne^{ik_x x + ik_y y}[a_q \vec{h}_q e^{ik_z z} e^{-i\omega t} + a_q^\dagger \vec{h}_q^* e^{-ik_z z} e^{i\omega t}], \quad (2)$$

where $N$ is the normalized factor, $\vec{k}_\parallel = (k_x, k_y)$ is the propagation vector in the $x$-$y$ plane. The parameters $a_q^\dagger$ and $a_q$ are the creation and annihilation operators, respectively. The parameters $\vec{e}_q$ and $\vec{h}_q$ denote the electromagnetic field polarization vectors, and $\omega$ is the angular frequency.

Choose $\vec{k}_\parallel$ to be parallel to the $x$ axis, the wave vector can be written as $\vec{k} = (k_x, k_y, k_z) = K_0(\alpha, 0, \gamma)$, and $K_0 = \omega/c$, $K_\parallel = K_0\alpha$, $K_\perp = K_0\gamma$. Consider the case that two plates are electrically anisotropic but magnetically isotropic, the permittivity and permeability tensors of the plates can be expressed as

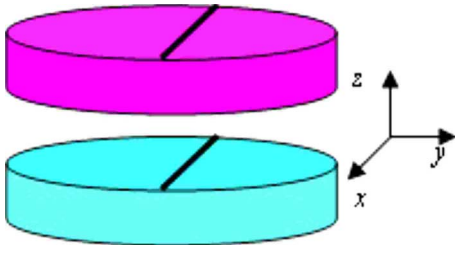



FIG. 1. (Color online) Two uniaxially parallel plates with diameter $D$ and the thickness $d$ in vacuum. The optic axes of the two plates are parallel to the $x$ direction.

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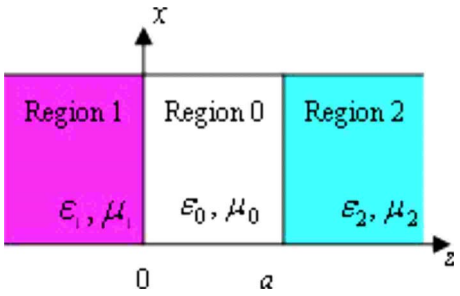



FIG. 2. (Color online) The schematic configuration of vacuum space within two plates. The optical property of the region $i$ is described by corresponding permittivity $\varepsilon_i$ and permeability $\mu_i$ ($i=0,1,2$).

$$\varepsilon_i = \begin{pmatrix} \varepsilon_{i,\parallel} & 0 & 0 \\ 0 & \varepsilon_{i,\perp} & 0 \\ 0 & 0 & \varepsilon_{i,\perp} \end{pmatrix}, \quad \mu_i = \begin{pmatrix} \mu_i & 0 & 0 \\ 0 & \mu_i & 0 \\ 0 & 0 & \mu_i \end{pmatrix}, \quad (i=1,2), \tag{3}$$

where the subscripts $\parallel$ and $\perp$ indicate the tensor component with a direction parallel or perpendicular to the optical axis, respectively. Hence, the Casimir force per unit area between the plates can be expressed as

$$F(a) = -\frac{\hbar}{2\pi^2 c^3}\int_1^\infty p^2 dp \int_0^\infty \xi^3 d\xi \times \left[\left(\frac{T_1+\mu_1 p}{T_1-\mu_1 p}\frac{T_2+\mu_2 p}{T_2-\mu_2 p}e^{2\xi p a/c}-1\right)^{-1} + \left(\frac{S_1+\varepsilon_{1,\parallel} p}{S_1-\varepsilon_{1,\parallel} p}\frac{S_2+\varepsilon_{2,\parallel} p}{S_2-\varepsilon_{2,\parallel} p}e^{2\xi p a/c}-1\right)^{-1}\right], \tag{4}$$

where $T_i=\sqrt{p^2-1+\mu_i\varepsilon_{i,\perp}}$, $S_i=\sqrt{\frac{\varepsilon_{i,\parallel}}{\varepsilon_{i,\perp}}(p^2-1)+\mu_i\varepsilon_{i,\parallel}}$, $i=1,2$, $K_\parallel^2=\frac{\xi^2}{c^2}(p^2-1)$, and $\omega=i\xi$.

Equation (4) has a similar form as Lifshitz's original expression [29], and it could be reduced to Lifshtz's expression if taking $\mu_i=\mu_0=1$ and $\varepsilon_{i,\parallel}=\varepsilon_{i,\perp}$.

A variable $\delta_i$, defined as $\varepsilon_{i,\parallel}=(1+\delta_i)\varepsilon_{i,\perp}=(1+\delta_i)\varepsilon_i$, is introduced to describe the degree of anisotropy of the plate $i$, then Eq. (4) could be rewritten as

$$F(a) = -\frac{\hbar}{2\pi^2 c^3}\int_1^\infty p^2 dp \int_0^\infty \xi^3 d\xi \times \left[\left(\frac{s_1+\mu_1 p}{s_1-\mu_1 p}\frac{s_2+\mu_2 p}{s_2-\mu_2 p}e^{2\xi p a/c}-1\right)^{-1} + \left(\frac{s_1+\varepsilon_1\sqrt{1+\delta_1}p}{s_1-\varepsilon_1\sqrt{1+\delta_1}p}\frac{s_2+\varepsilon_2\sqrt{1+\delta_2}p}{s_2-\varepsilon_2\sqrt{1+\delta_2}p}e^{2\xi p a/c}-1\right)^{-1}\right], \tag{5}$$

where $s_i=\sqrt{p^2-1+\mu_i\varepsilon_i}$, $i=1,2$.

The dispersion behavior of permittivity and permeability of materials are usually described by the Drude-Lorentz model [30]

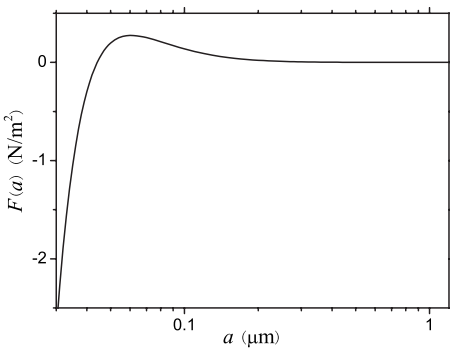



FIG. 3. The Casimir force between a gold plate and an anisotropic gold-based metamaterial plate as a function of the separation $a$. The positive (negative) value corresponds to the repulsive (attractive) force. $\Omega_p=\Omega=1.38\times10^{16}$ rad/s is the gold plasma frequency [3]. Parameters used are $\Omega_D=\Omega$, $\gamma_m=\gamma_e=0.005\Omega$, $\gamma_D=\gamma=0.004\Omega$, $\Omega_e=0.04\Omega$, $\Omega_m=0.009\Omega$, $\omega_e=\omega_m=0.1\Omega$, $f_2=3\times10^{-5}$, $\delta_2=0.001$.

$$\varepsilon(i\xi) = 1+\frac{\Omega_{pe}^2}{\omega_e^2+\xi^2+\gamma_e\xi} \tag{6}$$

and

$$\mu(i\xi) = 1+\frac{\Omega_{pm}^2}{\omega_m^2+\xi^2+\gamma_m\xi}. \tag{7}$$

In these representations, $\Omega_{pe}$ ($\Omega_{pm}$) is the plasma frequency or electric (magnetic) oscillating strength, $\omega_e$ ($\omega_m$) is the electric (magnetic) resonance frequency, and $\gamma_e$ ($\gamma_m$) is the electric (magnetic) dissipation factor. For metals, it becomes Drude model [29]

$$\varepsilon(i\xi) = 1+\frac{\Omega_p^2}{\xi^2+\gamma\xi}, \quad \mu(i\xi)=1. \tag{8}$$

However, for metallic-based metamaterial, the permittivity is usually considered to be a usual Drude-Lorentz response with a small Drude background [17,24,25]

$$\varepsilon(i\xi) = 1+(1-f)\frac{\Omega_e^2}{\omega_e^2+\xi^2+\gamma_e\xi}+f\frac{\Omega_D^2}{\xi^2+\gamma_D\xi}, \tag{9}$$

where $\Omega_D$ and $\gamma_D$ are Drude parameters of the metal in metamaterial, and the fill factor $f$ represents the fraction of metal in the metamaterial.

The Casimir force between plates made by various materials can be calculated using Eqs. (5)–(9). Figure 3 shows the Casimir force between a metallic plate and a metallic-based metamaterial plate with a fixed $\delta_2$ (the parameters are provided in the caption).

Figure 3 shows that the Casimir force is attractive for separation less than 0.01 $\mu$m. It decreases gradually as the separation goes larger. For a specific separation, named "transition separation" or "zero-force position," the force becomes zero. Crossing over this separation, the force turns to be repulsive. One plate with large permittivity and another with large permeability are required [9–11] to obtain a repulsive force. In other words, the magnetic effect of the plates

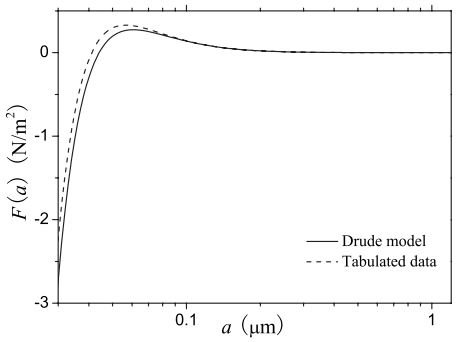



FIG. 4. The Casimir force between a gold plate and an anisotropic gold-based metamaterial plate. The dash line represents the result using tabulated optical data to describe the permittivity of gold. The solid line is the result shown in Fig. 3.

results in the repulsive Casimir force. If the permittivity and permeability are both constants [15,16], the response of the plates to all modes between them will be the same. In this case, the Casimir force will be always either repulsive or attractive and the attractive-repulsive transition can not occur. When the permittivity and permeability are functions of frequency, the response of the plates to different mode depends on the separation between the plates. For a small separation, the active modes are mainly high frequency modes in which the magnetic response of plates is negligible. Thus the electric effect becomes dominant and results in an attractive Casimir force. As the separation is getting larger and crossing over the zero-force position, low frequency modes, in which the plates have strong magnetic response, become more significant. Therefore, the Casimir force turns to be repulsive.

The reliability of our calculated results can be confirmed by recalculating the force using the tabulated optical data. The permittivity of gold is calculated using the optical tabulated data [31] with the help of dispersion relation [30]. The result represented in Fig. 4 shows a good agreement between the calculations using tabulated data and Drude model.

Since most metamaterial are anisotropic, we are interested in how the degree of anisotropy $\delta_1$ affects both the polarity and the magnitude of the Casimir force. The force-separation curve with different $\delta_2$ and the force-$\delta_2$ curve with a fixed separation is shown in Fig. 5.

Figure 5(b) shows an attractive-repulsive transition of Casimir force and it is decreasing almost linearly with increasing $\delta_2$. For the case of a metal plate and a metamaterial one, we can expand Eq. (5) with small $\delta_2$

$$F(a) = F_0(a) + F_\delta(a), \tag{10}$$

$$F_0(a) = -\frac{\hbar}{2\pi^2 c^3}\int_1^\infty p^2 dp \int_0^\infty \xi^3 d\xi \times \left[\left(\frac{s_1+\mu_1 p}{s_1-\mu_1 p}\frac{s_2+\mu_2 p}{s_2-\mu_2 p}e^{2\xi pa/c}-1\right)^{-1} + \left(\frac{s_1+\varepsilon_1 p}{s_1-\varepsilon_1 p}\frac{s_2+\varepsilon_2 p}{s_2-\varepsilon_2 p}e^{2\xi pa/c}-1\right)^{-1}\right], \tag{11}$$

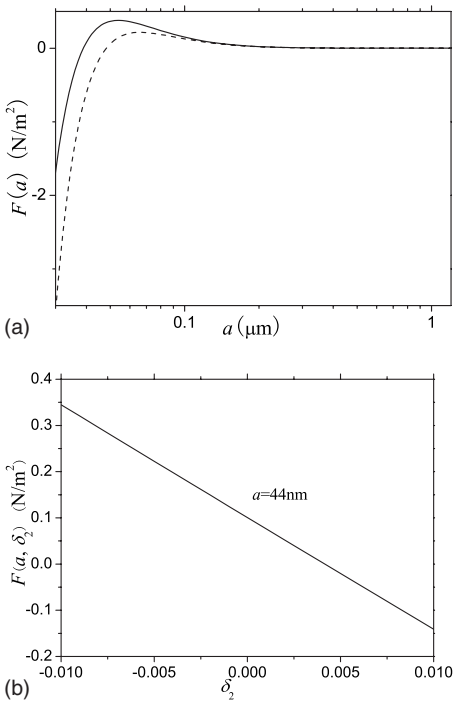



FIG. 5. (a) The Casimir force between a gold plate and an anisotropic gold-based metamaterial plate is plotted as a function of the separation $a$. The solid line is obtained with $\delta_2=-0.01$ and the dash line is obtained with $\delta_2=0.01$. The other parameters are the same as those used in Fig. 3(b). The Casimir force between a gold plate and an anisotropic gold-based metamaterial plate is plotted as a function of $\delta_2$ with $a=44$ nm.

$$F_\delta(a) = -\delta_2\frac{\hbar}{2\pi^2 c^3}\int_1^\infty p^2 dp \int_0^\infty \xi^3 d\xi \times \left[-\frac{s_1-\varepsilon_1 p}{s_1+\varepsilon_1 p}\frac{\varepsilon_2 s_2 p e^{2\xi pa/c}}{(s_2-\varepsilon_2 p)} \times \left(\frac{s_1+\varepsilon_1 p}{s_1-\varepsilon_1 p}\frac{s_2+\varepsilon_2 p}{s_2-\varepsilon_2 p}e^{2\xi pa/c}-1\right)^{-2}\right], \tag{12}$$

where $F(a)$ is the Casimir force between isotropic materials and $F_\delta(a)$ is the contribution of $\delta_2$, the degree of anisotropy of plate 2.

Equations (10)–(12) show that $F(a)$ is linear with respect to a small $\delta_2$ when the separation $a$ is fixed. Moreover, at $F_0(a)=0$, the polarity of Casimir force is determined by $\delta_2$. For a separation around $F_0(a)=0$, the attractive-repulsive transition will happen as $\delta_2$ changes. This result is consistent with previous numerical calculation.

According to the previous calculations, the transition separation is so small and the slope of the transition curve is so sharp, such that it is difficult to be observed in experiments. However, if we make $\varepsilon_1$ and $\varepsilon_2$ ($\mu_1$ and $\mu_2$) closer and still keep $\varepsilon_1>\varepsilon_2$ ($\mu_1<\mu_2$), a larger transition separation can be obtained. This may be easily carried out by changing the parameters of the metamaterial in Eqs. (7) and (9). Figure 6 shows the Casimir force between two metamaterial plates. The transition-separation is about 325 nm. Around this sepa-

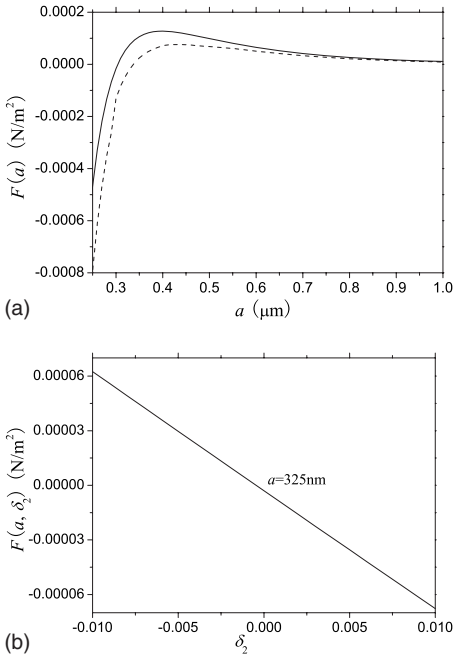



FIG. 6. (a) The Casimir force between two anisotropic gold-based metamaterial plates versus $a$. The solid line is calculated with $\delta_2=-0.01$ and the dash line is calculated with $\delta_2=0.01$. The parameters are $\Omega=1.38\times10^{16}$ rad/s, for plate 1, $\Omega_{D1}=\Omega$, $\gamma_{m1}=0.005\Omega$, $\gamma_{D1}=\gamma_{e1}=0.004\Omega$, $\Omega_{e1}=0.04\Omega$, $\Omega_{m1}=0.01\Omega$, $\omega_{e1}=0.5\Omega$, $\omega_{m1}=0.1\Omega$, $f_1=0.008$, $\delta_1=0.001$; for plate 2, $\Omega_{D2}=\Omega$, $\gamma_{D2}=0.004\Omega$, $\gamma_{m2}=\gamma_{e2}=0.005\Omega$, $\Omega_{e2}=0.04\Omega$, $\Omega_{m2}=0.1\Omega$, $\omega_{e2}=0.5\Omega$, $\omega_{m2}=0.1\Omega$, $f_2=2.7\times10^{-5}$. (b) The Casimir force between two anisotropic gold-based metamaterial plates separated 325 nm versus $\delta_2$.

ration, the polarity of Casimir force varies with both the separation and the degree of anisotropy. Therefore, it might possibly be observed by using atomic force microscope [8].

To calculate the thermal Casimir force, we can just replace the integration over $\xi$ by a summation by means of $\int d\xi/2\pi\rightarrow(k_BT/\hbar)\Sigma_m$ with $\xi\rightarrow\xi_m$ in Eq. (5),

$$F(a)=-\frac{k_BT}{\pi c^3}{\sum_m}'\xi_m^3\int_1^\infty p^2dp \times\left[\left(\frac{s_1+\mu_1p}{s_1-\mu_1p}\frac{s_2+\mu_2p}{s_2-\mu_2p}e^{2\xi pa/c}-1\right)^{-1} +\left(\frac{s_1+\varepsilon_1\sqrt{1+\delta_1}p}{s_1-\varepsilon_1\sqrt{1+\delta_1}p}\frac{s_2+\varepsilon_2\sqrt{1+\delta_2}p}{s_2-\varepsilon_2\sqrt{1+\delta_2}p}e^{2\xi pa/c}-1\right)^{-1}\right], \tag{13}$$

where the prime means that the $m=0$ term is counted with half weight, $k_B$ is Boltzmann constant, and $T$ is the temperature.

Although there have been controversies about the usage of Drude model in the calculations of thermal Casimir effect [32–35], we will still use Drude model for metals, since it

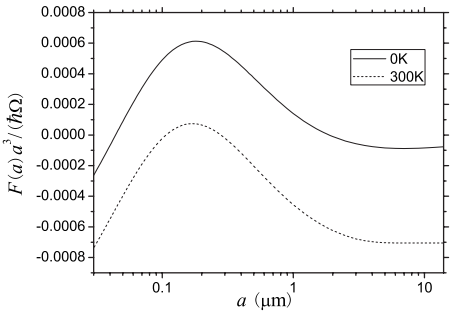



FIG. 7. Finite temperature Casimir effect between a gold plate and an anisotropic gold-based metamaterial plate versus the separation $a$. The dash line corresponds to $T=300$ K, while the solid line corresponds to $T=0$ K. The parameters are the same as that used in Fig. 3.

can describe the metal property very well in a wider frequency range [31]. In this case, we have $\xi_m^2\varepsilon(i\xi_m)\rightarrow0$ when $m=0$, hence

$$\frac{s_1-\mu_1p}{s_1+\mu_1p}=\frac{\sqrt{c^2K_\parallel^2+\xi_m^2\varepsilon_1}-\mu_1\sqrt{c^2K_\parallel^2+\xi_m^2}}{\sqrt{c^2K_\parallel^2+\xi_m^2\varepsilon_1}+\mu_1\sqrt{c^2K_\parallel^2+\xi_m^2}}\xrightarrow{m=0}0 \tag{14}$$

for metal plate ($\mu_1=1$), which means that the TE mode has no contribution to the $m=0$ term.

According to our numerical calculation as shown in Fig. 7, the finite temperature effect seems to reduce the strength of repulsion. This is because the $m=0$ term, which is dominant at 300 K, always contributes to attractive force if the Drude mode is used. The result agrees with the earlier study of Henkel *et al.* [36].

Recently, Milonni and collaborators published their work [17,24] on the Casimir force of metamaterials using the scattering approach [37]. They use two different fill factors in different directions to describe the degree of electric anisotropy. Alternatively, we consider this problem simply by introducing one parameter $\delta$ only, and give the direct expression of the anisotropy dependence of Casimir force in case of small $\delta$. Besides, the direction of uniaxial anisotropy of the plates is different in these two works. They focus on the out-of-plane condition in which the optic axes are perpendicular to the plates surface. We consider the in-plane case, in which more complicated phenomena are expected to occur [14–16,18], although we only discuss the case of parallel optic axes here. However, in both works, the attractive-repulsive transition can happen, which shows agreement and, at least to a certain level, the equivalence between the quantized surface mode technique and the scattering method.

In conclusion, we calculate the Casimir force between a metallic plate and an anisotropic metamaterial plate as well as the case for two anisotropic metamaterial plates. The Casimir force is attractive for a small separation and turns to be repulsive as the separation increases. Near the zero-force separation, the polarity of Casimir force depends very sensi-

tively on the degree of anisotropy, such that the attractive-repulsive transition can be observed while the degree of anisotropy changes. For present technology, performing an accurate quantitative experiment might still be difficult. The relevant effects, like the unparallel of the optical axes, will be discussed in the future.

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